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Effects of Stress and Friction Heterogeneity on Spatiotemporal Complexity of Seismic and Aseismic Slip on Strike-Slip Faults

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Key Points:

- Creep patches within seismogenic zone can generate stress heterogeneity and variable hypocenter depths in 2D models of strike-slip faults
- We find that the ratio of seismogenic zone width to nucleation size is not the only factor controlling slip complexity
- Models using slip law produce fewer partial ruptures, smaller stress drops, and lower peak slip rates compared to models using aging law

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Numerical and laboratory models of earthquake cycles on faults governed by rate-and-state friction often show cycle-invariant behavior, while natural faults exhibit considerable variability in slip history. Possible explanations include heterogeneities in fault stress and frictional properties. We investigate how various types of heterogeneity in simulations of quasi-dynamic sequences of seismic and aseismic slip affect rupture complexity, hypocenter location, and slow slip events (SSEs). We model a 2D vertical strike-slip fault and study the roles of self-affine fractal heterogeneities in normal stress, rate-and-state parameter ($a - b$), and characteristic slip-weakening distance, as well as the effects of a low-rigidity fault zone. We find that only a combination of heterogeneous parameters introduces variability in the modeled rupture extent, hypocenter depth, and recurrence interval. In particular, variable hypocenter depths require creeping patches within the velocity-weakening seismogenic zone. A low-rigidity fault zone adds little to slip complexity. Slip law simulations produce fewer partial ruptures, smaller stress drops, and lower peak slip rates compared to aging law simulations. We show that the ratio of the seismogenic zone thickness to nucleation size does not entirely predict slip complexity. The most complex aging law model, combining multiple heterogeneities, features system-size earthquakes preceded by cascades of partial ruptures and spontaneous SSEs. For such models, a transition from aperiodic to quasi-regular regimes requires more cycles than is typically needed to erase the effect of initial conditions. These results highlight the importance of heterogeneity in reproducing natural fault slip complexity in numerical models of earthquake sequences.

Plain Language Summary Natural earthquakes exhibit considerable variability in their size, time, and location. In contrast, computer simulations of earthquake cycles often show cycle-invariant behavior with constant recurrence intervals and characteristic slip distributions. Possible model ingredients needed to reproduce the variable slip histories observed in nature include heterogeneities in fault properties. We investigate how various types of heterogeneity in simulations of sequences of earthquakes affect rupture complexity. We find that only a combination of multiple heterogeneities introduces variability in the modeled earthquake size, location, and recurrence interval. In particular, variable hypocenter depths require areas of stable creep within the seismogenic zone. Including a layer of less stiff material near the fault adds little to slip complexity. The most complex model in this study, combining multiple heterogeneities, features non-repeating cycles of large earthquakes preceded by a cascade of smaller foreshocks. However, the complexity transitions to periodic cycles after ∼1,700 years of simulation time, suggesting heterogeneity-induced variability in modeled slip history may only be transient. Our results also suggest that 3D effects may be important for producing and maintaining spatiotemporal complexity in fault slip.

1. Introduction

The theory of rate-and-state friction (RSF; Dieterich, 1979; Ruina, 1983) is widely used to model seismic and aseismic slip on geological faults (e.g., Barbot et al., 2012; Erickson et al., 2020; Jiang & Lapusta, 2016; Tse & Rice, 1986). Simulations of earthquake sequences on faults obeying RSF successfully reproduce many aspects of the observed behavior of natural faults, including stick-slip (e.g., Lapusta et al., 2000; V. C. Li & Rice, 1987), afterslip (Barbot et al., 2009; Helmstetter & Shaw, 2009; Perfettini & Avouac, 2007; K. Wang & Fialko, 2014), interseismic creep (Kaneko et al., 2013; Lindsey & Fialko, 2016), the Gutenberg-Richer and Omori laws (Beall et al., 2022; Cattania, 2019; Dieterich, 1994; Ito & Kaneko, 2023), and earthquake triggering (Luo & Liu, 2019; Perfettini et al., 2003a, 2003b; Wei et al., 2018).

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However, models of sequences of earthquakes and aseismic slip (SEAS) on faults governed by RSF often exhibit cycle invariance, whereby the recurrence intervals and slip patterns become constant following some initial "spin-up phase" in which the effects of initial conditions are erased (e.g., Erickson et al., 2020; Rice, 1993; Tse & Rice, 1986; S. Wang, 2024), particularly when no heterogeneity is assumed. Such models also exhibit a limited range of earthquake nucleation depths (e.g., Barbot, 2019; Cattania, 2019; Erickson et al., 2023; Lapusta & Rice, 2003). This is in contrast to the observed seismicity patterns on natural faults that are characterized by highly irregular recurrence intervals and a wide distribution of hypocenter locations throughout the seismogenic zone (e.g., Jin & Fialko, 2020; Ross et al., 2020; Waldhauser & Schaff, 2008).

Previous studies reproduced the observed complex slip behavior by introducing inherent discreteness, for example, by using numerical grids that are coarser than the characteristic nucleation size (e.g., Ben-Zion & Rice, 1995; Rice, 1993; Rice & Ben-Zion, 1996; Ziv & Cochard, 2006), or modifying constitutive parameters to facilitate nucleation (e.g., Cochard & Madariaga, 1996; Shaw & Rice, 2000). In continuum models, spatiotemporal complexity of slip can arise from various heterogeneities, including spatially variable frictional properties (Hillers et al., 2007; Jiang & Fialko, 2016; M. Li et al., 2025; Luo & Ampuero, 2018; Molina-Ormazabal et al., 2023), elastic moduli (Y. Huang et al., 2014; Idini & Ampuero, 2020; Thakur et al., 2020), and fault geometry (Cattania & Segall, 2021; Ozawa & Ando, 2021; Tal & Gabrieli, 2024; Yin et al., 2023). Models that allow for intermingled patches of stable and unstable slip have shown that interactions between these patches may also enhance slip complexity (J. Huang & Turcotte, 1990; Kato, 2020; Nakata et al., 2011) and produce realistic foreshock migration patterns and Omori-type aftershock sequences (Dublanchet et al., 2013; Ito & Kaneko, 2023; Luo & Ampuero, 2018). However, in many previous studies, the heterogeneities are assumed to be regularly spaced and piece-wise constant (e.g., Ito & Kaneko, 2023; Kato, 2020; Molina-Ormazabal et al., 2023) or randomly distributed (e.g., Galvez et al., 2020; Hillers et al., 2006), often with bimodal strength distributions. While the spatial distribution of subsurface frictional properties remains poorly constrained, a deeper understanding of how strongly varying heterogeneous fault properties affect the complexity of earthquake cycles is warranted.

Alternatively, it was shown that increases in slip complexity can result from decreases in the critical length scale of the nucleation process relative to the characteristic fault size (Cattania, 2019; Erickson et al., 2011; Herrendörfer et al., 2015; Y. Liu & Rice, 2007; Nie & Barbot, 2022). Numerical models show less regular cycles of earthquakes for smaller characteristic nucleation size, with the emergence of partial ruptures (e.g., Barbot, 2019; Cattania & Segall, 2019; Lapusta et al., 2000) and realistic earthquake statistics such as the Omori-type aftershock decay or the frequency-magnitude relation (e.g., Cattania, 2019). However, the respective models still exhibit a narrow range of hypocenter locations, mostly limited to the edges of the seismogenic zone (e.g., Barbot, 2019; Lapusta & Rice, 2003).

Complexity in rupture evolution in SEAS models can additionally be introduced by various other sources. For example, complex fault geometry, such as curved or rough fault surfaces (Ozawa & Ando, 2021; Perez-Silva et al., 2022; Yin et al., 2023) and effects of pore fluids (e.g., Zhu et al., 2020) have been shown to produce ruptures of variable sizes, varying nucleation locations, and aseismic transients. Incorporating full elastodynamics may facilitate more variable system dynamics, by accounting for the stress transfer via radiated seismic waves (e.g., Thakur et al., 2020). Introducing bulk inelasticity has been shown to produce complex rupture evolution (e.g., Abdelmeguid et al., 2024; Mia et al., 2023). However, many SEAS studies adopt a quasi-dynamic approach (Equation 2) with a simple, planar fault embedded in a purely elastic medium, omitting fluid interactions. In this study, we seek to identify mechanisms that can generate considerable complexity in models that admit several simplifying assumptions. We discuss potential limitations of omitting physically relevant factors such as inertia (Sections 4.1 and 4.5).

The largely empirical RSF framework involves a choice of the functional form of the so-called evolution law for the state variable (e.g., Ampuero & Rubin, 2008; Dieterich, 1979; Ruina, 1983). The two most common choices are the aging law (Dieterich, 1979) and the slip law (Ruina, 1983), but other forms of the evolution law were proposed as well (e.g., Kato & Tullis, 2001; Linker & Dieterich, 1992; Nagata et al., 2012; Sato et al., 2025; Yoshida et al., 2020). Different formulations are meant to explain different aspects of available experimental data. For example, the aging law well captures the time-dependent healing of the rock surface in slide-hold-slide and stick-slip experiments (e.g., Beeler et al., 1994; Dieterich & Kilgore, 1994; Mitchell et al., 2015, 2016), while the slip law seemingly better accounts for the evolution of friction in velocity stepping experiments with large

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velocity changes (e.g., Ampuero & Rubin, 2008; Bayart et al., 2006; Pignalberi et al., 2024). These empirical laws have different intrinsic length scales and predict different slip evolutions away from the steady state (Ampuero & Rubin, 2008). How various choices of the state evolution law may affect the complexity of the system behavior and a fault's response to external stress perturbations is not well understood.

In this study, we investigate the effects of various types of heterogeneity on rupture complexity, hypocenter location, and aseismic transients in simulated earthquake and aseismic slip sequences. SEAS simulations empowered by high-performance computing enable more realistic parameterization, extensive parameter space exploration, and volume-discretized methods, addressing key knowledge gaps despite high computational costs (e.g., Erickson & Dunham, 2014; D. Liu et al., 2020; Pranger, 2020; Taufiqurrahman et al., 2023; Thakur et al., 2020; Uphoff et al., 2023). We perform a suite of 2D quasi-dynamic SEAS simulations on a planar vertical strike-slip fault under the antiplane strain approximation, incorporating different heterogeneities on and off the fault (Sections 3.1 and 3.3) and using two different state variable evolution laws (Section 3.4). We document key features of the most complex models obtained in this study, including cascades of ruptures, emergence of spontaneous slow slip events, and quasi-chaotic earthquake sequences (Section 3.2). We find that heterogeneity in a single parameter is insufficient to sustain complex rupture behavior in a 2D quasi-dynamic framework. Moreover, models with comparable ratios of the seismogenic fault width to characteristic nucleation length scale can exhibit markedly different levels of system complexity (Section 4.2). Diversity in hypocenter depths is strongly affected by the presence of creeping patches within the seismogenic zone, and less so by a low-rigidity fault zone. These results provide insights into how the system behavior, including the rupture characteristics, depends on various heterogeneities in material properties and field variables, as well as different assumed evolution laws. In a companion paper, we use the complex multi-cycle simulations developed here, incorporating heterogeneous material properties, to investigate the triggering potential of rate-and-state friction faults perturbed by static and dynamic stress changes from nearby earthquakes (Yun et al., 2025a).

2. Methods

2.1. Models of Earthquake Sequences on Faults Governed by RSF

We use the open-source code *Tandem* (Uphoff et al., 2023) to perform 2D antiplane strain quasi-dynamic SEAS simulations on a vertical strike-slip fault embedded in an elastic halfspace (Figure 1a). *Tandem* is based on a symmetric interior penalty discontinuous Galerkin method and is optimized for high-performance computing. *Tandem* uses the regularized version of the rate-and-state friction formulation (Lapusta et al., 2000):

$$F(\|\boldsymbol{V}\|,\theta) = a\sinh^{-1}\left[\frac{\|\boldsymbol{V}\|}{2V_0}\exp\left(\frac{f_0 + b\,\ln(V_0\theta/D_{RS})}{a}\right)\right], \tag{1}$$

where F is the instantaneous friction coefficient, $\|\boldsymbol{V}\|$ is the Euclidean norm of the slip rate vector $\boldsymbol{V}$, θ is the state variable, a, b are the rate-and-state parameters for direct and evolution effect, respectively, D_{RS} is the characteristic state evolution distance, V_0 is the reference slip rate, and f_0 is the reference friction coefficient. All SEAS model parameters used in this study are summarized in Table 1.

The sign of $(a - b)$ determines the stability of the system. An increase in sliding velocity leads to a drop of static friction when $a - b < 0$, promoting instability, which is referred to as velocity-weakening (VW) behavior. Conversely, static friction increases when $a - b > 0$, suppressing instability, which is defined as velocity-strengthening (VS) behavior. In this study, we include shallow and deep VS regions surrounding a central VW zone, representing a 10-km-wide seismogenic zone (Figure 1a). The rate-and-state fault is loaded from the bottom creeping zone and the far boundary with a constant velocity (V_{pl}) of 10^{-9} m/s ≈30 mm/yr, corresponding to the long-term fault slip rate.

In quasi-dynamic simulations, the inertial effect is approximated by a radiation damping term $\eta\boldsymbol{V}$ (Rice, 1993):

$$-\boldsymbol{\tau} = \sigma_n\, F(\|\boldsymbol{V}\|,\theta)\frac{\boldsymbol{V}}{\|\boldsymbol{V}\|} + \eta\boldsymbol{V}, \tag{2}$$

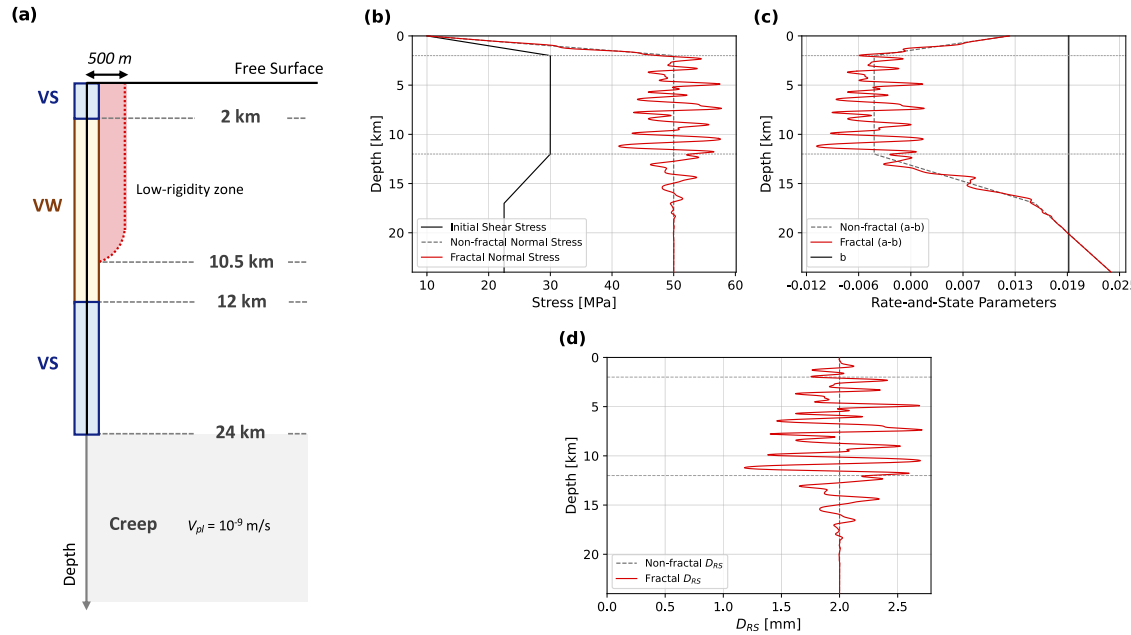


Figure 1. (a) Sketch of the model geometry for the SEAS simulations using *Tandem*. The rate-and-state fault (black vertical line) includes a central velocity-weakening zone (yellow) surrounded by shallow and deep velocity-strengthening zones (blue). The bottom creep zone governed by the constant loading rate (V_{pl}) is shaded in gray. The red-shaded area indicates the spatial extent of a low-rigidity fault zone. As the model represents a perfectly symmetric vertical strike-slip fault, we model only one side of the domain. (b–d) Self-affine fractal distributions of (b) effective normal stress (version 3), (c) rate-and-state parameters (version 1), and (d) characteristic state evolution distance (version 1). The fractal distributions of all three parameters share the same limiting wavelengths of $\lambda_{\mathrm{min}} = 500$ m and $\lambda_{\mathrm{max}} = 2.5$ km. See Table S1 in Supporting Information S1 for a detailed description of each fractal model.

where $\eta = \mu/2c_s$ is half of the shear-wave impedance with shear modulus μ and shear-wave speed c_s, and $\boldsymbol{\tau}$ and σ_n are shear and normal stresses on the fault, respectively. Although quasi-dynamic models do not capture all aspects of fully dynamic solutions, particularly in the presence of geometric or rheological barriers (e.g., Kroll et al., 2023; Lambert & Lapusta, 2021), they produce qualitatively comparable slip patterns at considerably reduced computational cost (Lapusta & Rice, 2003; Rice & Ben-Zion, 1996; Thomas et al., 2014). We discuss the potential effect of the quasi-dynamic approximation in more detail in Section 4.1.

Table 1
Base Values for the Parameters Used in the SEAS Models Using Tandem

Symbol	Parameter	Value
a	Rate-and-state parameter, direct effect	Varies (see Figure 1c)
b	Rate-and-state parameter, evolution effect	0.019
D_{RS}	Characteristic state evolution distance	4 mm
f_0	Reference coefficient of friction	0.6
V_0	Reference slip rate	10^{-6} m/s
V_{init}	Initial slip rate	10^{-9} m/s
V_{pl}	Plate loading rate	10^{-9} m/s
σ_n	Background effective normal stress	50 MPa (see Figure 1b)
τ^0	Background shear stress	Varies (see Figure 1b)
ν	Poisson's ratio	0.25
W	Seismogenic zone width[a]	~10 km
L_f	Rate-and-state fault length	24 km

[a]May slightly vary due to fractal heterogeneity (see Section 2.2).

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The evolution of the state variable θ is governed by an ordinary differential equation. We explore the two most common end-members, the aging law (Dieterich, 1979):

$$\frac{d\theta}{dt} = 1 - \frac{\|\boldsymbol{V}\|\theta}{D_{RS}}, \tag{3}$$

and the slip law (Ruina, 1983):

$$\frac{d\theta}{dt} = -\frac{\|\boldsymbol{V}\|\theta}{D_{RS}} \ln\left(\frac{\|\boldsymbol{V}\|\theta}{D_{RS}}\right). \tag{4}$$

We set the initial value for the state variable (θ^0) to be consistent with the background shear stress (τ^0) and initial slip rate (V_{init}):

$$\theta^0 = \frac{D_{RS}}{V_0} \exp\left\{\frac{a}{b} \ln\left[\frac{2V_0}{V_{\text{init}}} \sinh\left(\frac{\tau^0 - \eta V_{\text{init}}}{a\sigma_n}\right)\right] - \frac{f_0}{b}\right\}. \tag{5}$$

The ratio of the length of the seismogenic fault (W; Table 1) to the critical nucleation size is known to affect the complexity of the earthquake sequence. For 2D antiplane simulations using the aging law (with $0.5 < a/b < 1$), the critical nucleation size (L_∞) can be expressed as follows (Rubin & Ampuero, 2005):

$$L_\infty = \frac{2}{\pi} \frac{\mu b D_{RS}}{\sigma_n (b-a)^2}. \tag{6}$$

There is no equivalent analytical form of the critical nucleation size for the slip law, as the nucleation zone continuously shrinks under the slip law formulation (Ampuero & Rubin, 2008). Thus, we only compare the W/L_∞ ratios among aging law models. The critical nucleation size also controls the grid size, as the former needs to be resolved by the model. A detailed description of different model resolution requirements for each evolution law is provided in Text S1 in Supporting Information S1.

We use adaptive time stepping handled by the software PETSc (Abhyankar et al., 2014; Balay et al., 1997, 2019) with a fourth-order embedded fifth-order Dormand-Prince scheme Runge-Kutta method.

2.2. Fractal Heterogeneities

As one realization of the poorly constrained distribution of heterogeneity in natural fault zone properties, we assume a self-affine fractal variation that captures the scale-dependent variability observed, for example, in fault roughness (Candela et al., 2012; Lee & Bruhn, 1996; Maurer, 2024; Renard et al., 2006). We introduce band-limited self-affine fractal variations to the effective normal stress (σ_n), rate-and-state parameters ($a - b$), and the characteristic state evolution distance (D_{RS}). The 1D fractal distributions are characterized by the power spectral density $P(k)$ as follows (Andrews & Barall, 2011; Dunham et al., 2011):

$$P(k) \propto k^{-(2H+1)} \tag{7}$$

with the wavenumber k and the Hurst exponent H. The Hurst exponent $H = 1$ results in a self-similar fractal distribution, while $0 \leq H < 1$ produces a self-affine distribution. For natural faults, H is typically assumed to vary between 0.4 and 0.8 (Renard & Candela, 2017). We assume $H = 0.7$ for most of the fractal profiles used in this study (Cattania & Segall, 2021). We also evaluate the effects of higher values of the Hurst exponent ($H = 1.9$) on rupture evolution (see Text S3 in Supporting Information S1). The fractal variation is limited between minimum (λ_{min}) and maximum (λ_{max}) wavelengths. We explore a wide range of λ_{min} from 25 m (comparable to the critical nucleation size) to 750 m and λ_{max} from 2.5 km to 10 km (comparable to W) to identify values of λ_{min} and λ_{max} that produce most diversity in both rupture sizes (e.g., occurrence of both partial ruptures and system-size events) and hypocenter depths (i.e., widely distributed nucleation locations within the seismogenic zone).

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We use a Fourier transform method (Andrews & Barall, 2011; Shi & Day, 2013) to generate the fractal profile and take an amplitude-to-wavelength ratio $\alpha = 10^{-2}$ to scale the root-mean-square amplitude of the profile (Dunham et al., 2011). Since our model fault is 1D, the α value refers to variations in the direction perpendicular to slip. All fractal variations are tapered outside the seismogenic zone by scaling their amplitude by the distance from the nearest VW region. The fractal amplitudes are then converted into variations of parameters by applying scaling factors that match the order of magnitude of each parameter (Table S1 in Supporting Information S1). For example, the fractal effective normal stress profile is obtained by scaling the fractal amplitude by $(\rho_c - \rho_w)g$, where $\rho_c = 2670$ kg/m^3 is density of crust, $\rho_w = 1000$ kg/m^3 is density of water, and $g = 9.8$ m/s^2 is the acceleration due to gravity. Since the fractal heterogeneity has a mean of zero, the average value for each parameter (i.e., $\overline{\sigma_n}$, $\overline{a - b}$, and $\overline{D_{RS}}$) remains the same for both fractal (red solid lines in Figures 1b–1d) and non-fractal (gray dashed lines in Figures 1b–1d) distributions. All fractal heterogeneity models explored in this study are summarized in Table S1 in Supporting Information S1 and shown in Figure S1 in Supporting Information S1.

Our models do not explicitly account for fault roughness or fault geometric effects (e.g., Cattania & Segall, 2021; Ozawa & Ando, 2021; Tal & Gabrieli, 2024), but instead prescribe heterogeneous stress and strength conditions on a planar fault. These may differ from rough-fault models where shear drag can preserve heterogeneity throughout the cycle (Dunham et al., 2011; Shi & Day, 2013). Given our 2D antiplane strain model assumptions, incorporating true geometric roughness is beyond the scope of this study.

2.3. Event Detection and Classification

We implement an automated event detection and classification algorithm to systematically compare the event time and hypocenter locations across different models. A seismic event is identified when the peak slip rate along the fault exceeds a threshold velocity, which varies between 0.01 m/s and 0.2 m/s, for longer than 0.5 s at more than one of the evaluation points, which are spaced every 200 m along the fault. An event is disregarded if the difference between the maximum and minimum peak slip rates during the event is less than 15% of the threshold velocity to eliminate minor fluctuations in slip rate. A detailed description of the post-processing procedures is provided in Text S2 in Supporting Information S1.

A "system-size earthquake" is defined as an event that ruptures a length greater than 10 km (i.e., the entire seismogenic zone), while all other events are denoted "partial rupture events" hereafter. A "leading foreshock" is defined as the first partial rupture event in a sequence that eventually leads to a system-size earthquake.

2.4. Assessing the Level of Complexity

To quantitatively evaluate the degree of complexity across different models, we introduce a "complexity score" based on three aspects of slip history: (a) how widely the hypocenters of system-size earthquakes are distributed as a function of depth, (b) the periodicity of the sequence, and (c) whether partial ruptures exist and if so, how diverse their sizes are. Each aspect is scored between 0 and 1, and the combined complexity score, the sum of all three metrics, ranges from 0 to 3.

The complexity score for hypocenter depth heterogeneity of system-size earthquakes is computed based on the histogram of hypocenter depths of system-size earthquakes. We count the number of bins n along depth (0.5 km interval) that contain at least one system-size earthquake hypocenter. To ensure a score of zero when all events occur at the same depth, we subtract 1 from the bin count $(n - 1)$. The respective score is then $(n - 1)/(N - 1)$, where N is the maximum value of n in all of the tested models.

The complexity score for periodicity is computed from the standard deviation of indices for the nearest system-size events with hypocenter depths less than 0.5 km apart. This standard deviation is normalized to 1 to provide the final score.

Lastly, the complexity score for heterogeneity in partial rupture events is computed from the standard deviation of rupture lengths among partial rupture events. If no partial rupture events are present, the score is assigned a value of zero, and the resulting score is normalized to 1.

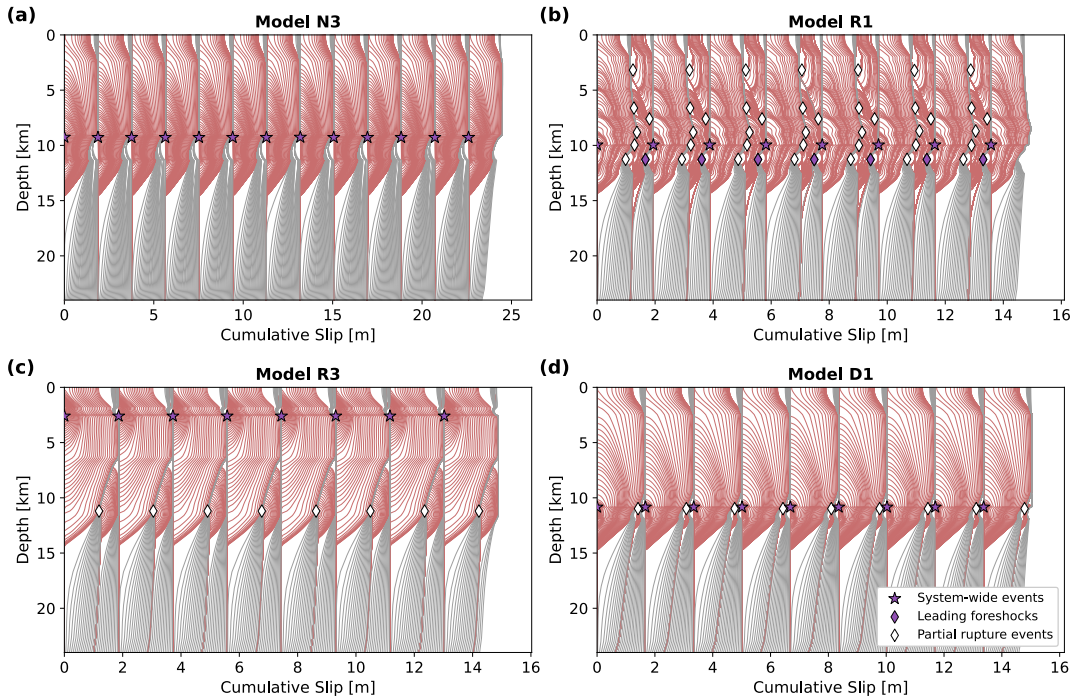


Figure 2. Cumulative slip evolution along the fault in exemplary SEAS simulations with heterogeneity in either stress or strength. (a) Model with heterogeneity in effective normal stress shown in Figure 1b (model N3). (b–c) Models with heterogeneity in $(a - b)$ parameter (b) with (model R1) and (c) without (model R3) the velocity-strengthening patches embedded within the seismogenic layer. Model R1 in panel (b) uses the $(a - b)$ heterogeneity shown in Figure 1c. (d) Model with heterogeneity in D_{RS} parameter shown in Figure 1d (model D1). All models show the cumulative slip omitting the first 200 yr of spin-up time. Pink contours, drawn every 0.5 s, show the coseismic evolution of slip, while gray contours, plotted every 2 yr, show the longer-term evolution of slip. Purple stars, purple diamonds, and white diamonds indicate the hypocenter locations of system-size earthquakes, leading foreshocks, and partial rupture events, respectively.

3. Results

3.1. Models With Heterogeneity in a Single Parameter

We explore a range of heterogeneities to yield a SEAS model with realistic variability in both event size and hypocenter depth distribution. The W/L_∞ ratio in these models varies with depth, due to the fractal spatial distribution of parameters. This study covers the range of maximum W/L_∞ values between 5 and 400 (Figure 9). All SEAS models explored in this study are summarized in Table S2 in Supporting Information S1.

Using the aging law, we find that neither heterogeneity in a single parameter (Figures 2 and 3) nor the presence of a low-rigidity fault zone (Figure 5) alone is sufficient to introduce the desired complexity. Most of the models with heterogeneity in a single parameter exhibit characteristic cycles and hypocenters located only at the periphery of the seismogenic zone, similar to results from models that do not assume any heterogeneity (e.g., Lapusta & Rice, 2003; Lindsey & Fialko, 2016). We also tested models in which normal stress increases with depth with superimposed fractal heterogeneity (Figure 3c), but the cycles remained repeatable, with nucleation limited to the bottom edge of the seismogenic zone, where the critical nucleation size (L_∞, Equation 6) is the smallest. It is expected that depth-dependent normal stress favors nucleation at the base of the seismogenic zone but adds little to complexity (e.g., Zhang et al., 2006).

Two notable exceptions to the general lack of complexity in single-heterogeneity models are a model with a variable $(a - b)$ profile featuring VS-VW transitions (model R1; Figure 2b) and a model with heterogeneous σ_n with a larger amplitude-to-wavelength ratio, α (model N7; Figure 3d). Introducing an $(a - b)$ profile with VS patches within the seismogenic VW region gives rise to earthquakes that nucleate at various depths within the seismogenic zone, rather than only at its periphery. Earthquakes nucleate at the boundaries of VS patches, where the stressing rate is increased due to creep on VS patches. Nucleation in the middle of the seismogenic zone is

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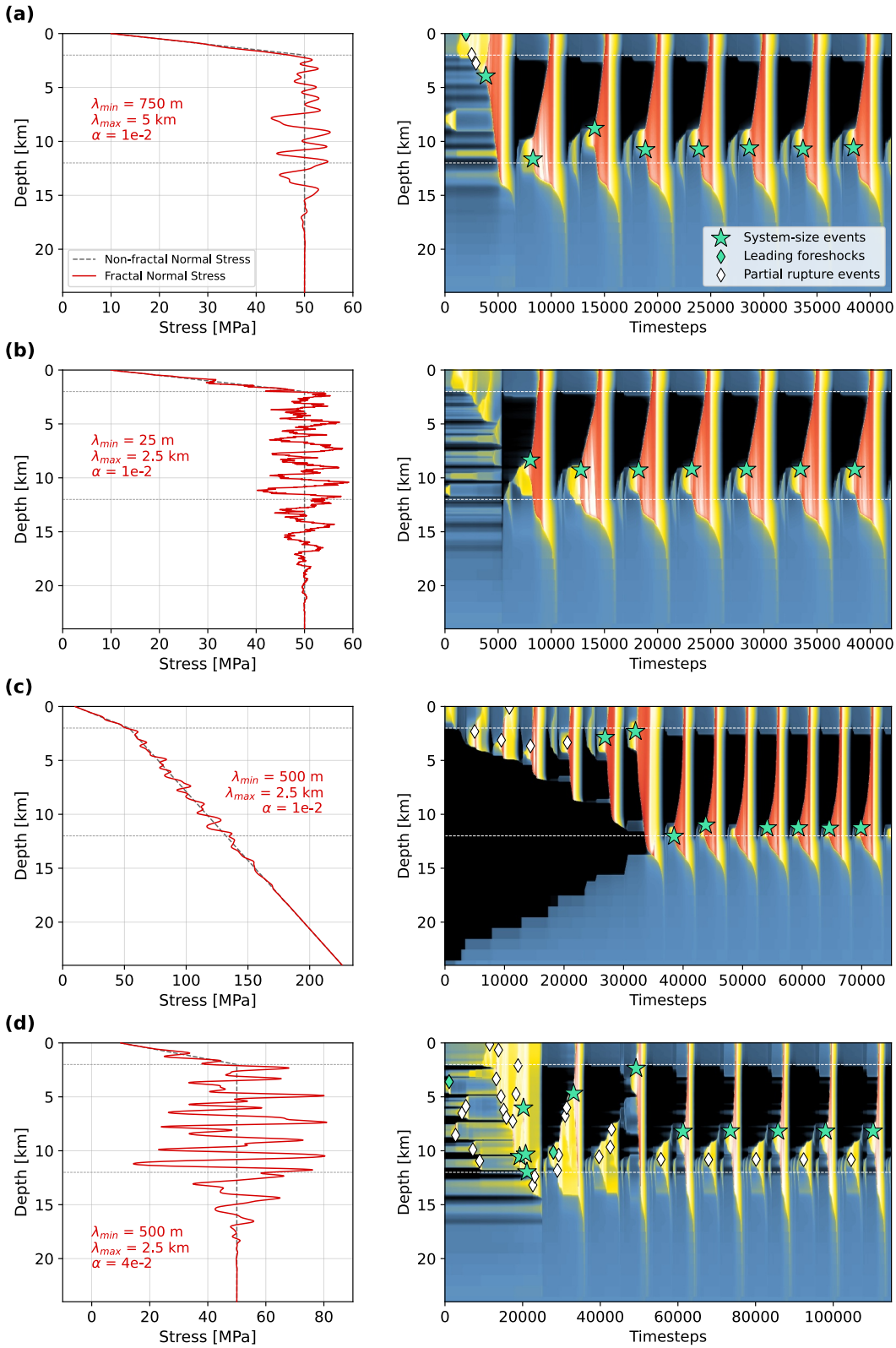


Figure 3.

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absent in other heterogeneous $(a - b)$ profiles that lack VS patches embedded within the VW layer (model R3; Figure 2c).

In model N7, we use a larger $\alpha = 4 \times 10^{-2}$, increasing the variability in the normal stress heterogeneity. The root-mean-square normal stress variation $\Delta\sigma_{\mathrm{rms}}$ in this model is ~10 MPa, compared to ~3 MPa in other models with heterogeneous σ_n. This elevated heterogeneity leads to considerably more complex rupture evolution, featuring partial ruptures, creep patches within the seismogenic zone, and spontaneous shallow and deep slow slip events (SSEs) at its base. Importantly, earthquakes in model N7 nucleate within the seismogenic zone (Figure 3d), unlike any other models with heterogeneous σ_n. Creep patches begin to emerge during the later stages of the earthquake cycle, preceding the next system-size event, and are colocated with regions of moderately low effective normal stress (20–40 MPa). SSEs occur within the patch of the lowest σ_n (~10 MPa; see also Section 4.6). The SSEs in model N7 load a nearby high-σ_n patch, triggering partial ruptures that in turn load another high-σ_n patch. A system-size earthquake occurs when the patch with the highest normal stress becomes sufficiently stressed by preceding SSEs and partial ruptures. These results may be comparable to those of previous models analyzing heterogeneous normal stress induced by rough surfaces and how this promotes complex rupture sequences (Cattania & Segall, 2021; Tal et al., 2018).

The lack of complexity in single-heterogeneity models is likely not related to the choice of Hurst exponent. We test two different values of H (0.7 and 1.9) and find only minor effects on rupture evolution in models with heterogeneity in D_{RS} (group D) and normal stress (group N), including minimal changes in recurrence intervals and unchanged complexity scores. The effect of different Hurst exponents is more pronounced in models with heterogeneity in $(a - b)$ (group R). A detailed discussion is provided in the Text S3 in Supporting Information S1.

3.2. Models With Combined Stress and Frictional Heterogeneities

Combining an $(a - b)$ profile that includes VS patches within the seismogenic zone with heterogeneity in other model parameters, for example, models N3R1, N3R1D1, and N3R2D1, consistently produces cycles with system-size earthquakes preceded by a cascade of partial ruptures (Figures 4 and 6). The hypocentral depths of the system-size events are broadly distributed throughout the seismogenic zone. The maximum W/L_∞ values range from 60 to 400 (Figure 9). Nonetheless, all these models remain quasi-periodic, with a constant nucleation depth for all system-size earthquakes in each model, respectively, although the nucleation depth may be different in different models.

When heterogeneity in all three parameters ($(a - b)$, D_{RS}, and σ_n) is combined, and the $(a - b)$ profile has large amplitude variations (as in model N3R2D1), spontaneous deep and shallow SSEs are generated (Figure 7), consistent with observations (Rousset et al., 2019; Vavra et al., 2024; Wei et al., 2009). Deep SSEs in model N3R2D1 spatially coincide with a small VW patch embedded within the deep VS zone. This suggests that instability is initiated at the VW patch but fails to grow into a runaway seismic rupture due to the VS barriers located above and below. Shallow SSEs occur only after a sequence of partial rupture events, presumably to relax the stress induced by the preceding sequence. The peak slip rate of the shallow SSEs is an order of magnitude lower than that of the deep SSEs (Figures 7a and 7b). Both shallow and deep SSEs are often followed by a sequence of partial rupture events, similar to the observation of aseismic slip preceding small to moderate earthquakes (e.g., L. Huang et al., 2024; Linde et al., 1988; Thurber, 1996; Thurber & Sessions, 1998). A detailed discussion of the mechanism underlying SSE generation in our models is provided in Section 4.6.

We do not find a comparable level of complexity in models without heterogeneous $(a - b)$ profiles with VS patches embedded within the seismogenic zone. For example, model N3D1 (Figure 5b) shows a relatively simple rupture evolution, in which initial heterogeneities are rapidly wiped out, approaching a periodic cycle consisting

Figure 3. Comparison of SEAS models with heterogeneity only in effective normal stress (group N): models (a) N1 ($\alpha = 10^{-2}$, $\lambda_{\mathrm{min}} = 750$ m, $\lambda_{\mathrm{max}} = 5$ km), (b) N4 ($\alpha = 10^{-2}$, $\lambda_{\mathrm{min}} = 25$ m, $\lambda_{\mathrm{max}} = 2.5$ km), (c) Nd1 (depth-dependent; $\alpha = 10^{-2}$, $\lambda_{\mathrm{min}} = 500$ m, $\lambda_{\mathrm{max}} = 2.5$ km), and (d) N7 ($\alpha = 4 \times 10^{-2}$, $\lambda_{\mathrm{min}} = 500$ m, $\lambda_{\mathrm{max}} = 2.5$ km). The left column shows the spatial distribution of the normal stress heterogeneity, and the right column shows the resultant spatiotemporal evolution of slip rate. Green stars, green diamonds, and white diamonds indicate the hypocenter locations of system-size earthquakes, leading foreshocks, and partial rupture events, respectively. Numbers next to the markers indicate the event number. α: amplitude-to-wavelength ratio; λ_{min} & λ_{max}: minimum and maximum wavelengths of the fractal variation.

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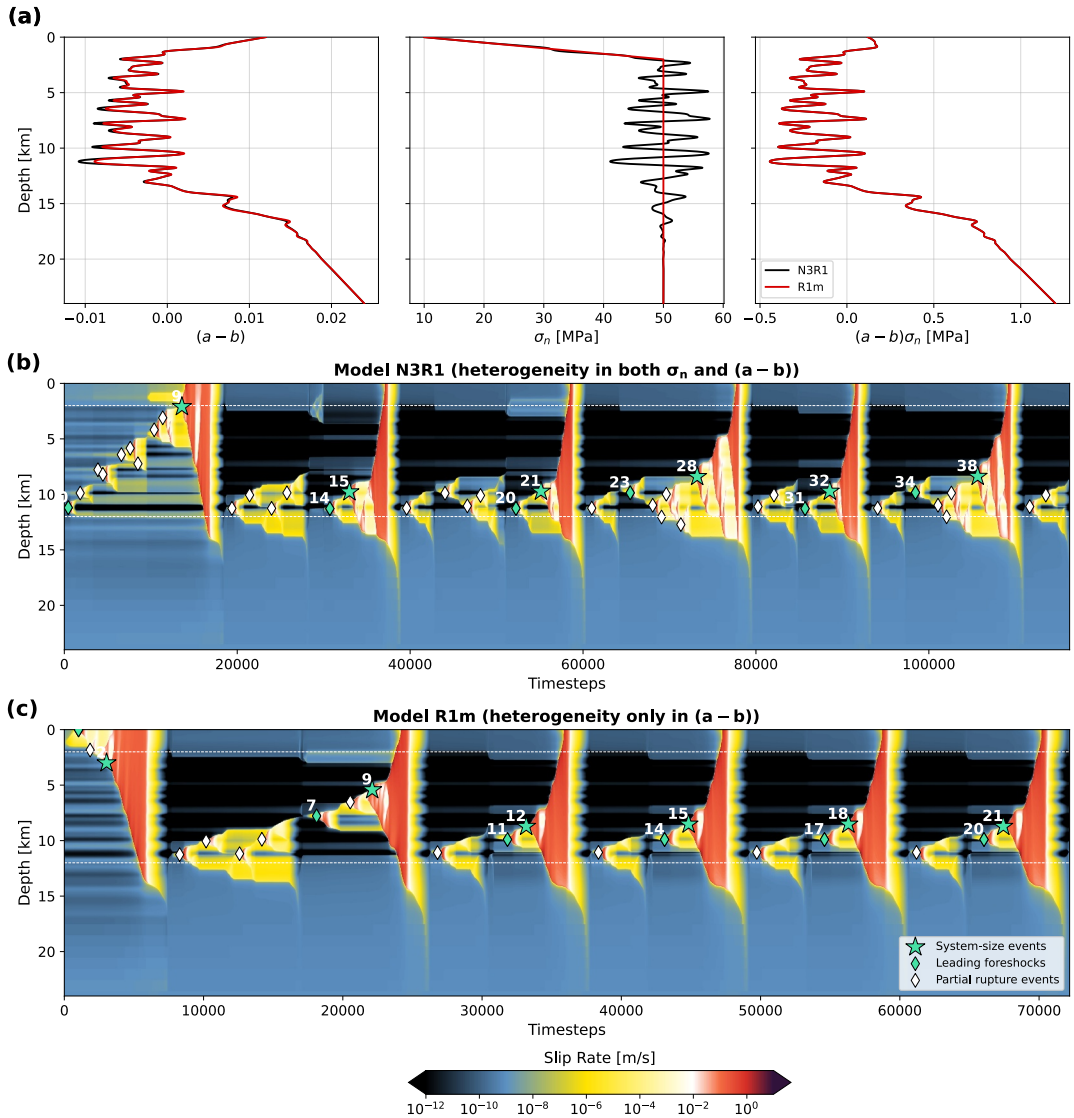


Figure 4. Comparison of model N3R1 and model R1m. (a) Spatial distribution of $(a - b)$, σ_n, and the resultant frictional strength $\sigma_n(a - b)$ used in models N3R1 (black) and R1m (red). (b–c) Spatiotemporal evolution of slip rate during 700-year simulations of (b) model N3R1 and (c) model R1m, showing less complex slip evolution in model R1m. Green stars, green diamonds, and white diamonds indicate the hypocenter locations of system-size earthquakes, leading foreshocks, and partial rupture events, respectively. Numbers next to the markers indicate the event number.

of a single partial rupture and a system-size event. This result illustrates the importance of sustained stress heterogeneity, achieved either through alternating rheological properties (e.g., VS-VW transitions) or pronounced heterogeneity (e.g., model N7), which creates variability in nucleation locations and rupture size. A more detailed discussion of the driving mechanisms controlling variations in hypocentral depth is provided in Section 4.1.

Models with combined heterogeneities in multiple parameters exhibit more complex rupture patterns compared to those with heterogeneity in only one parameter, even when frictional strength, defined as $\sigma_n(a - b)$, is the same. For example, models N3R1 and R1m share the same spatial distribution of frictional strength. However, model N3R1 includes correlated fractal heterogeneity in both σ_n and $(a - b)$ (black lines in Figure 4a), whereas model R1m has heterogeneity only in $(a - b)$ that matches the frictional strength profile of model N3R1 (red lines in Figure 4a). Note that the individual σ_n and $(a - b)$ profiles differ between models N3R1 and R1m, even though both are constructed to match the same $\sigma_n(a - b)$. If frictional strength alone governed system complexity, and

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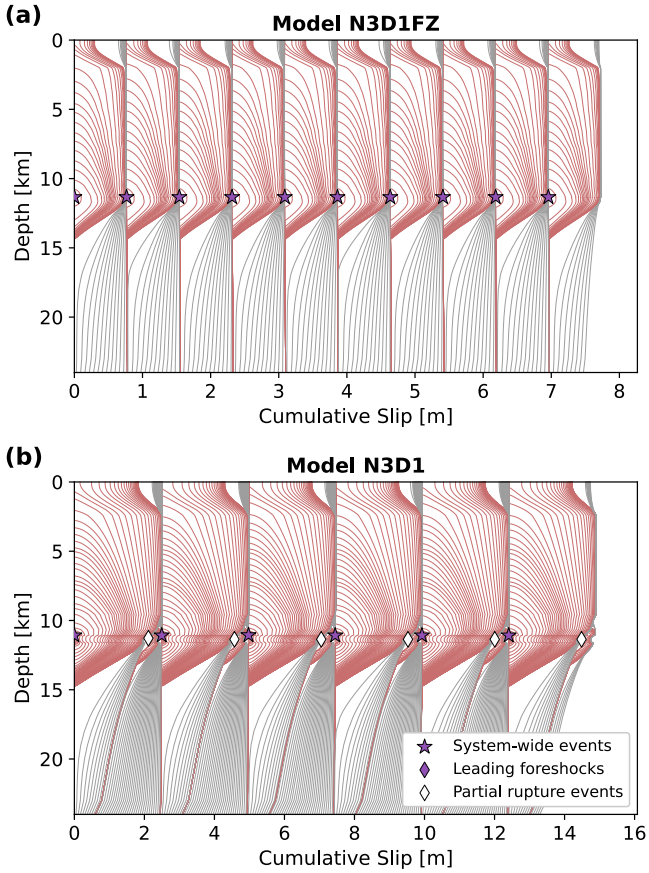


Figure 5. Comparison of SEAS models (a) with and (b) without a low-rigidity fault zone surrounding the fault (red shaded area in Figure 1a). (a) Model N3D1FZ, featuring a low-rigidity fault zone ($\mu_{FZ} = 20$ GPa) embedded in the bulk with $\mu = 32$ GPa. (b) Model N3D1, featuring a uniformly low $\mu = 20$ GPa in the entire bulk. Both models share fractal heterogeneities in the effective normal stress and characteristic state evolution distance shown in Figures 1b and 1d. All models show the cumulative slip, omitting the first 200 yr of spin-up time. Pink contours, drawn every 0.5 s, show the coseismic evolution of slip, while gray contours, plotted every 2 yr, show the longer-term evolution of slip. Purple stars, purple diamonds, and white diamonds indicate the hypocenter locations of system-size earthquakes, leading foreshocks, and partial rupture events, respectively.

the contributions of σ_n and $(a - b)$ could not be distinguished, the two models should evolve into an identical SEAS cycle. Despite this matched frictional strength, model N3R1 produces more variable hypocenter depths of system-size events and more variable sizes of partial ruptures compared to model R1m (Figures 4b and 4c). The corresponding complexity scores (Section 2.4) are 0.7 for model N3R1 and 0.3 for model R1m, confirming that model N3R1 is more complex than model R1m. We assess differences between the two models only considering spun-up cycles, for example, after event 20 in model N3R1 (Figure 4b) and after event 11 in model R1m (Figure 4c). The persistence of this difference, even after the spin-up phase, indicates that the presence of a spatial stress gradient is important for producing complex rupture patterns in our models. This result is consistent with the increased complexity in model N7, which involves a larger gradient in normal stress.

Through trial-and-error, we identify one model (N12R4D4) that shows a longer, sustained aperiodicity (Figure 6c). In this model, system-size earthquakes are consistently preceded by a cascade of partial rupture events. Spontaneous SSEs emerge at shallow and deep edges of the seismogenic zone. This model incorporates heterogeneity in both $(a - b)$ and D_{RS} with a high Hurst exponent of 1.9. The heterogeneous normal stress profile in model N12R4D4 resembles a band-limited fractal distribution with $H = 1.9$ within a limited bandwidth (500 m–2.5 km), but contains a much richer short-wavelength component, compared to a strictly self-affine fractal distribution (gray vs. purple lines in Figure S6 in Supporting Information S1). The transition from fractal to non-fractal scaling occurs at about $\lambda_{\min} = 500$ m, which is comparable to the observationally defined crossover length scale, below which the log-linear scaling of the power spectra of a rough surface becomes unreliable (Candela et al., 2012). Therefore, we take this model as representing heterogeneity with larger roughness at short length scales. A detailed discussion of the longer-lasting complexity shown in model N12R4D4 is provided in Section 4.5.

3.3. Models With a Low Rigidity Fault Zone

We perform additional SEAS simulations adding a low-rigidity region surrounding the fault, as an analogy to damage zones developing near active faults (e.g., Chester et al., 1993; Fialko et al., 2002; Gabriel et al., 2024; Y. Huang et al., 2014). We include a 500 m wide and 10 km deep rectangular low-rigidity fault zone (with rigidity μ_{FZ}), which tapers at the bottom end as a semi-circle with a 500 m radius (Figure 1a).

We find that the presence of a low-rigidity fault zone has minimal impact on earthquake sequences across all considered cases in this study, which explores two different rigidity contrast values (30% and 60%; Table S2 in Supporting Information S1). Without the inclusion of fractal heterogeneities in the initial dynamic parameters (model FZ), the low-rigidity fault zone alone results in partial rupture events and system-size earthquakes, but the sequence remains cycle-invariant, with hypocenters located only at the bottom of the seismogenic zone.

Including a low-rigidity fault zone in models with fractal heterogeneities adds little to the complexity. For example, models N3D1FZ and N3D1 share the same fractal heterogeneity, but model N3D1FZ includes a low-rigidity fault zone ($\mu_{FZ} = 20$ GPa & $\mu = 32$ GPa), while model N3D1 has a uniformly low $\mu = 20$ GPa in the entire bulk. Both models exhibit periodic cycles without variability in hypocenter depth (Figure 5). While model N3D1 produces partial ruptures, model N3D1FZ shows only system-size events. Similar results are observed using a different Hurst exponent. Using $H = 1.9$, models N12D4FZ and N12D4 exhibit a comparable rupture evolution, characterized by periodic cycles consisting of a partial rupture and a system-size rupture nucleating at the base of the seismogenic zone (Figure S2 in Supporting Information S1).

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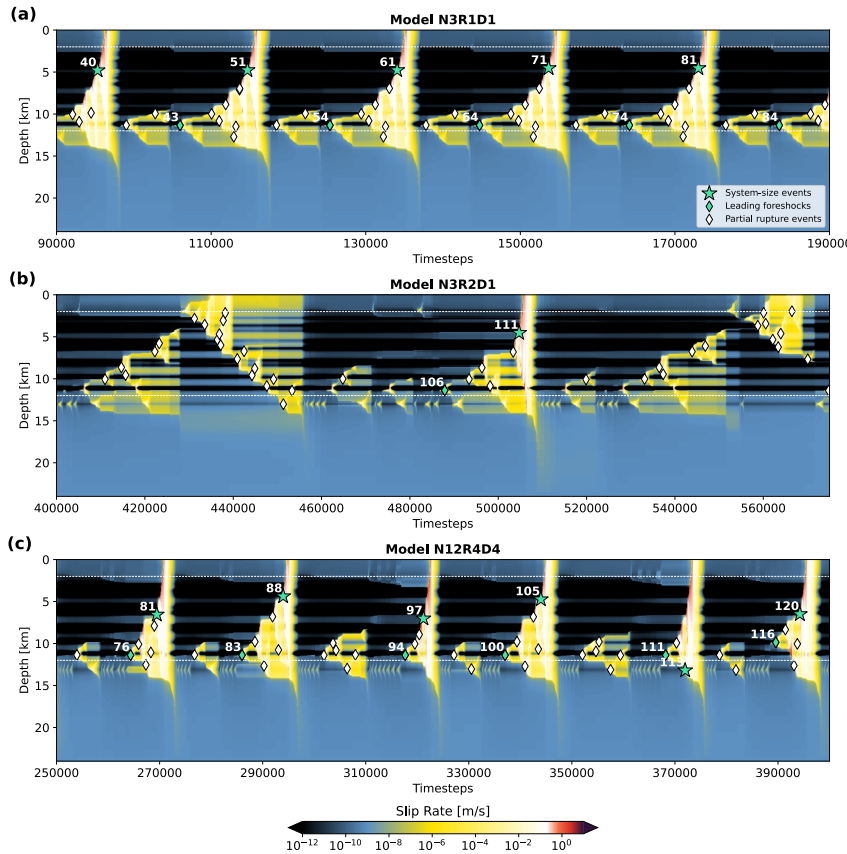


Figure 6. Spatiotemporal evolution of slip rate of SEAS models with heterogeneity in all three parameters (see Section 3.2). The slip rate evolution of (a) model N3R1D1 between 348 yr and 679 yr of simulation time, (b) model N3R2D1 between 387 yr and 537 yr of simulation time, and (c) model N12R4D4 between 745 yr and 1,180 yr of simulation time. Green stars, green diamonds, and white diamonds indicate the hypocenter locations of system-size earthquakes, leading foreshocks, and partial rupture events, respectively. Numbers next to the markers indicate the event number.

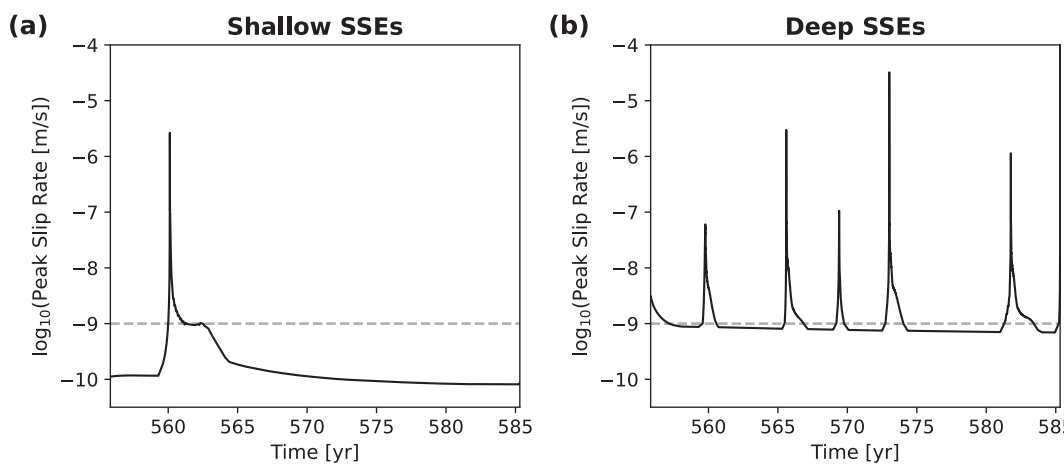


Figure 7. Slow slip events (SSEs) observed in model N3R2D1. (a, b) Peak slip rates of the shallow (<5 km) and the deep (10–20 km) SSEs. Gray dashed lines mark the constant plate loading rate (V_{pl}).

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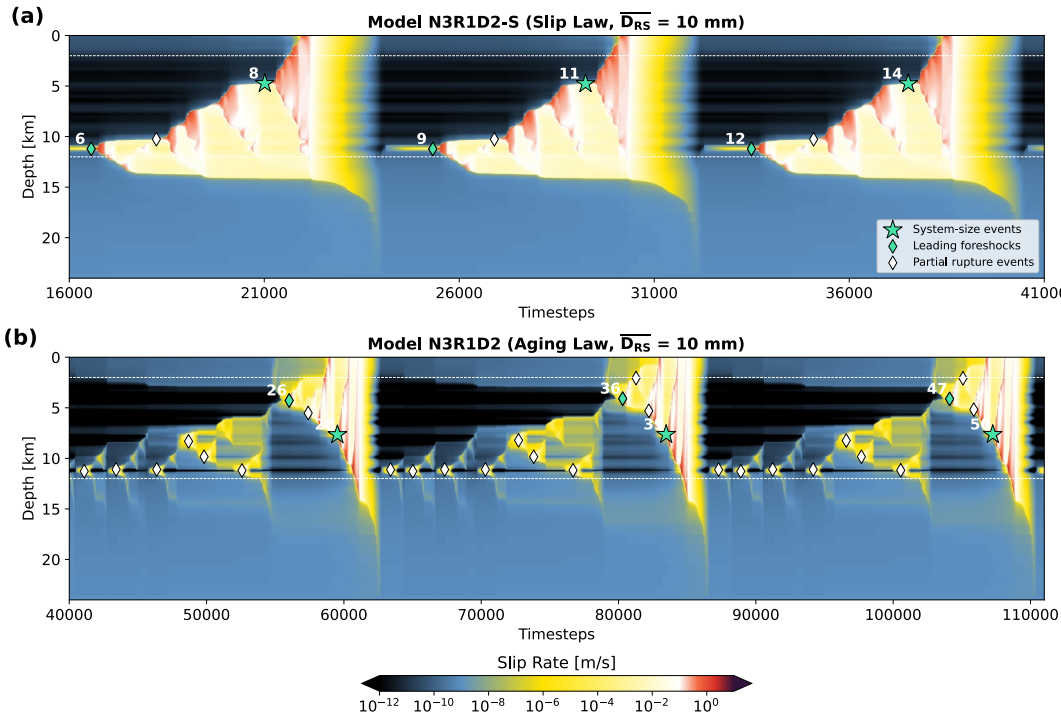


Figure 8. Comparison of spatiotemporal evolution of slip rate of models using (a) slip law (model N3R1D2-S) between 131 yr and 291 yr of simulation time and (b) aging law (model N3R1D2) sharing the same input parameters (see Section 3.4), between 262 yr and 613 yr of simulation time. Green stars, green diamonds, and white diamonds indicate the hypocenter locations of system-size earthquakes, leading foreshocks, and partial rupture events, respectively. Numbers next to the markers indicate the event number.

3.4. Comparing Aging Law and Slip Law Models

We perform a suite of simulations to assess the effect of using different evolution laws. The model assuming slip law is shown in Figure 8a. This model uses the same set of parameters as used in the N3R1D1 simulation but with an increased $\overline{D_{RS}}$ of 10 mm (D_{RS} heterogeneity version 2; Table S1 in Supporting Information S1) to reduce the computational burden (see Text S1 in Supporting Information S1). The modeled earthquake sequence (denoted as N3R1D2-S, where "S" stands for slip law) is characterized by the repetition of a few partial rupture events at the base of the seismogenic zone, followed by a system-size earthquake in the middle of the seismogenic zone (~5 km depth). Periodic earthquake sequences and a scarcity of smaller earthquakes are observed in model N3R1D2-S, as noted in previous studies that considered the effects of slip law (e.g., Rice & Ben-Zion, 1996; Rubin, 2008).

We produce a new aging law model using $\overline{D_{RS}}$ = 10 mm (denoted model N3R1D2; Figure 8b), for direct comparisons among different evolution laws. Models N3R1D2 and N3R1D1 differ only in the magnitude of $\overline{D_{RS}}$ (10 mm vs. 2 mm).

Model N3R1D2 shows more complex earthquake sequences with multiple partial rupture events preceding system-size earthquakes compared to the "equivalent" slip law model (N3R1D2-S). In model N3R1D2, multiple partial rupture events are connected by a prolonged aseismic slip, forming a sequence that eventually leads to a system-size earthquake. Due to this prolonged aseismic slip, each sequence in model N3R1D2 lasts for about 21 min on average, much longer compared to ~21 s in the N3R1D2-S model.

We find that the aging law (model N3R1D2) tends to produce larger static stress drops and peak slip rates during its system-size events compared to the slip law (model N3R1D2-S), although their differences are rather small: model N3R1D2 and the N3R1D2-S model have average stress drops of ~5 MPa and ~3 MPa, respectively, and average peak slip rates of ~3.6 m/s and ~3.4 m/s. Differences in stress drops and peak slip rates between models using aging and slip laws are consistent with results from previous studies (Hawthorne & Rubin, 2013;

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He et al., 2003; Perfettini & Ampuero, 2008; Pignalberi et al., 2024); we provide a more detailed comparison in Section 4.4.

4. Discussion

4.1. Variability in the Hypocenter Depth

Contrary to the predictions of classic SEAS models on RSF faults, which limit earthquake nucleation to the edges of the VW layer representing the seismogenic zone (Lapusta & Rice, 2003; Rice, 1993; Tse & Rice, 1986), natural faults exhibit a wide range of hypocenter depths (Melgar et al., 2023; Ross et al., 2020; Waldhauser & Schaff, 2008). In particular, in California, most earthquakes nucleate in the middle, rather than at the edges, of the seismogenic zone (Jin & Fialko, 2020).

The models presented here replicate such diverse hypocenter depths when we introduce VS patches within the seismogenic zone (Figures 1c and 2b). This variability arises because stress concentrates at the VS-VW transitions due to creep in the VS patch (e.g., Lapusta & Rice, 2003; Lindsey & Fialko, 2016). These transitions between VS and VW also facilitate the spontaneous occurrence of deep SSEs. Although the alternation of creeping and locked regions in our models clearly creates stress heterogeneity and facilitates variability in hypocenter depth, not all VS-VW transitions accommodate nucleation. We do not observe a clear relationship between earthquake hypocenters and the size or strength of the VS patches (Figure S3 in Supporting Information S1). Some VS patches that do not host nucleation, for example, a VS patch at ∼5 km depth in model R1 (Figure S3a in Supporting Information S1), instead act as rupture barriers. These patches are often small, with resistances expected to be minor based on scaling relations for uniform VS patches suggested by Kaneko et al. (2010). These results suggest that VS patches with variable sizes and strengths may alter stress conditions more effectively than uniform VS patches. Under fractally varying $(a - b)$, even small or weak VS patches may play a non-negligible role.

Another case that produces variability in hypocenter depth arises when we use heterogeneous effective normal stress with large amplitude variations, as in model N7 (Figure 3d). In this model, system-size earthquakes nucleate in the middle of the seismogenic zone, where patches of elevated normal stress are progressively loaded by preceding SSEs or partial ruptures. This behavior contrasts with other models with heterogeneous σ_n, which fail to nucleate system-size earthquakes, even when stress heterogeneity is generated by partial ruptures (e.g., Figure 5b). These results suggest that elevated static strength is required to enable earthquake nucleation within the seismogenic zone, not only at its periphery. In model N7, the ratio of $\Delta\sigma_{\mathrm{rms}}$ to the background normal stress level (50 MPa) is about 0.5, which is lower than the threshold of 1.1 suggested by Cattania and Segall (2021) for promoting smaller earthquakes preceding system-size earthquakes. However, the large total range of normal stress variation of ∼70 MPa (spanning from ∼10 MPa to ∼80 MPa) in this model may not be realistic. While this model confirms the role of strong normal stress variability in generating variability in nucleation location, caution is warranted when extrapolating these results to real faults.

In the context of the 2D quasi-dynamic, antiplane strain models on a planar fault presented here, no heterogeneity is capable of producing a broad distribution of event hypocenters unless it involves creeping patches (model R1) or significantly increased effective normal stress (model N7). Previous 2D quasi-dynamic, antiplane strain models exhibit similar limitations, even in models that showed variable rupture extent and realistic frequency-magnitude relationships (e.g., Barbot, 2019; Cattania, 2019). In contrast, such variability in nucleation depths is frequently observed in SEAS models with varying assumptions. 3D models show earthquake nucleation within a central VW layer, equivalent to the seismogenic zone in our study, not only when heterogeneous fault friction is introduced (e.g., heterogeneous D_{RS}: Galvez et al., 2020; Hillers et al., 2006; heterogeneous $(a - b)$: Jiang & Lapusta, 2016) but also for relatively simple parameter setups (e.g., Lapusta & Liu, 2009). Complex fault geometries (e.g., curved fault, rough surface, multiple faults) can introduce variability in nucleation location, even without heterogeneous model parameters (e.g., Ozawa & Ando, 2021; Perez-Silva et al., 2022; Yin et al., 2023). Accounting for fluid flow and permeability evolution successfully produces earthquake swarms within the seismogenic zone as well as slow transients (Ozawa et al., 2024; Zhu et al., 2020). Incorporating inelasticity in bulk material may also introduce aperiodic cycles and clustering of earthquakes as localized plastic strain accumulation facilitates segmentation of ruptures (e.g., Abdelmeguid et al., 2024; Mia et al., 2023). The inclusion of dynamic wave propagation may introduce additional variability in nucleation locations. For example, while both Abdelmeguid et al. (2019) and Thakur et al. (2020) model 2D antiplane slip on a vertical strike-slip fault

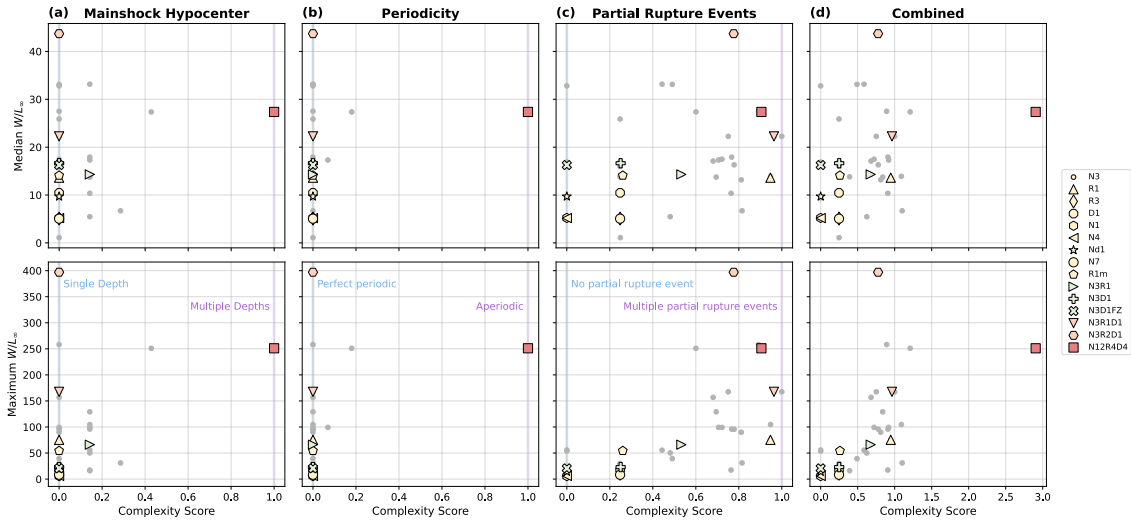


Figure 9. Correlation between the complexity score (Section 2.4) and W/L_∞ ratio for all aging law models (gray dots). The top and bottom rows show the median and maximum W/L_∞ ratios, respectively. (a–c) Individual complexity scores for (a) hypocenter depth heterogeneity of system-size earthquakes, (b) periodicity, and (c) heterogeneity in partial rupture events. (d) Combined complexity score, the sum of the three individual complexity scores, ranging from 0 to 3. Models presented in Figures 2–8 are plotted in colors and symbols, as shown in the legend. We colored single heterogeneity models in yellow, double heterogeneity models in light green, and triple heterogeneity models in light pink. The most complex model (N12R4D4) is colored in a dark pink color. See Table S2 in Supporting Information S1 for model name definition.

embedded in a low-rigidity fault zone, variability in hypocentral depths emerges only in the fully dynamic simulations of Thakur et al. (2020).

4.2. Control of W/L_∞ on the Slip Complexity

We systematically compare maximum and median values of W/L_∞ with the complexity scores (Section 2.4) across all aging law models (Figure 9). Overall, the maximum W/L_∞ ratio correlates well with the combined complexity score. A larger W/L_∞ ratio tends to increase the diversity in partial rupture events (Figure 9c), while having little impact on the periodicity (Figure 9b) and the hypocenter depth heterogeneity of system-size earthquakes (Figure 9a).

However, a larger W/L_∞ ratio does not always lead to a more complex model. For example, model D1 (octagons in Figure 9) has higher median and maximum W/L_∞ ratios than model N7 (circles in Figure 9). Yet, model D1 produces periodic cycles consisting of a partial rupture and a system-size earthquake that nucleate at a fixed depth, while model N7 produces variable hypocenter depths and SSEs. This lack of complexity in model D1, despite a higher W/L_∞ ratio compared to model N7, is likely related to the model setup differences. As discussed in Section 3.1, a large normal stress variation in model N7 promotes complex rupture sequences, which is not captured in the W/L_∞ ratio.

In addition, we evaluate whether the observed complexity stems from a high maximum W/L_∞ ratio or from heterogeneous fault properties. We perform a simulation with a constant high value of W/L_∞ of ~250, comparable to the maximum W/L_∞ value in model N12R4D4, but using homogeneous input parameters (Figure S4 in Supporting Information S1). This homogeneous model generates relatively simple, periodic cycles consisting of a system-size and a partial rupture, whose hypocenter depths are consistently located at the base of the seismogenic zone.

By systematically comparing W/L_∞ ratios using a quantitative metric (the complexity score), we find that the W/L_∞ ratio may not serve as an absolute criterion for predicting the degree of complexity among models with substantially different setups. Previous studies emphasizing the control of W/L_∞ ratio on complexity often utilize simple setups, such as depth-independent stress values (e.g., Cattania, 2019), or perform isolated parameter searches, such as changing the D_{RS} value while fixing other parameters (e.g., Barbot, 2019; Lapusta & Rice, 2003). These approaches offer systematic insights into how W/L_∞ affects rupture behavior under a simple, homogeneous setup. However, in models with pronounced heterogeneity, interaction between different forms of

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heterogeneity may additionally affect the system's complexity, diminishing the power of the W/L_∞ ratio for predicting the level of complexity across different models. This finding emphasizes that heterogeneity is driving complexity and can produce system behavior that is not easily inferable.

4.3. The Effect of a Low-Rigidity Fault Zone

We find that the presence of a low-rigidity fault zone does not increase the complexity of the earthquake cycle (e.g., compare models N3D1FZ and N3D1). This finding is in line with the comprehensive parameter search by Nie and Barbot (2022), who concluded that low-rigidity fault zones contribute to complexity only by decreasing the nucleation size. Models with and without such low-rigidity fault zones are indistinguishable when they share a similar W/L_∞ ratio.

The observed insensitivity of rupture behavior to the presence of a low-rigidity fault zone in our models may stem from the limited range of rigidity contrasts we explored. Nie and Barbot (2022), in a systematic study, showed that higher rigidity contrast tends to promote more complex ruptures at a fixed characteristic weakening distance (D_{RS}). Abdelmeguid et al. (2019) noted the emergence of partial ruptures as rigidity contrast increases. These findings suggest that a larger rigidity contrast may increase system complexity, possibly due to a reduction of nucleation size. Moreover, the effect of a low-rigidity fault zone may become increasingly important in models beyond pure elastic off-fault behavior, for example, those incorporating plasticity or viscoelasticity (e.g., Abdelmeguid & Elbanna, 2022; Erickson et al., 2017; Niu et al., 2024; van Dinther et al., 2014; Zhai et al., 2025; Zhao et al., 2024). However, it remains unclear whether incorporating higher rigidity contrasts or inelastic effects is sufficient to break the observed cycle-invariance.

The absence of complex wave interaction in our quasi-dynamic simulations may inhibit the emergence of complex rupture patterns, as pointed out by Nie and Barbot (2022). Previous fully dynamic models have shown changes in rupture behavior, such as a broader depth range of earthquake hypocenters and increased shear stress heterogeneity within the seismogenic zone, when low-rigidity fault zones are present (Y. Huang & Ampuero, 2011; Y. Huang et al., 2014; Thakur et al., 2020; Thakur & Huang, 2024). In this study, models with (N3D1FZ) and without (N3D1) the low-rigidity fault zone show comparable shear stress distributions 1 hr before and after a system-size earthquake (Figure S5 in Supporting Information S1).

4.4. Different Rupture Styles of the Aging Law and the Slip Law

In Section 3.4, we observe key differences between models using the aging law (model N3R1D2) and the slip law (model N3R1D2-S). The slip law model tends to produce: (a) quasi-periodic earthquake cycles of shorter duration, consisting of fewer earthquakes per cycle, (b) smaller average static stress drop, and (c) smaller peak slip rates during system-size earthquakes.

The rapid cascade of earthquake sequences with fewer smaller earthquakes in the slip law model is consistent with the highly non-linear nature of the slip law. The slip law shows slower growth of fracture energy during rupture acceleration, allowing instability to develop over a smaller length scale (i.e., smaller critical nucleation size) and facilitating easier rupture propagation across the entire fault (Ampuero & Rubin, 2008). Previous studies have shown that the slip law can accommodate unstable slip regardless of the stiffness of the system (Gu et al., 1984; Ranjith & Rice, 1999), particularly when subjected to stress perturbations (Helmstetter & Shaw, 2009; Perfettini & Ampuero, 2008).

While smaller static stress drops under the slip law are well-documented (Hawthorne & Rubin, 2013; He et al., 2003), peak slip rate comparisons between slip and aging laws have yielded varying conclusions across different models. For example, using the slip law, Perfettini and Ampuero (2008) reported higher peak slip rates under finite stress perturbation, while He et al. (2023) observed the opposite within a weakening zone during the nucleation process. These discrepancies may arise from differences in model setups, such as the spring-slider system or homogeneous frictional properties, since correlations between kinematic parameters may depend on the local heterogeneity of dynamic parameters (Schliwa et al., 2024; Schmedes et al., 2010; Vyas et al., 2023). To the best of our knowledge, this study presents the first systematic comparison of rupture characteristics for the aging and slip laws under more realistic conditions, incorporating fractal heterogeneity in stress and material properties.

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4.5. Transition From Aperiodic to Cycle-Invariant Regimes

We observe that most initial heterogeneities are rapidly wiped out after a few cycles (or a few hundred years), even when all three heterogeneous parameters are combined. Models then leave a so-called spin-up phase (Erickson et al., 2020; Hetland & Hager, 2006) and exhibit repeating slip sequences. Among all models explored in this study (Table S2 in Supporting Information S1), only model N12R4D4 sustains prolonged aperiodicity for more than 1,000 yr (Figure 6c). This may imply that when heterogeneities are either statistically unpredictable or composed of multiple uncorrelated components, the system requires more time to evolve toward (quasi-)cyclic slip behavior.

Even for model N12R4D4, the slip sequence eventually evolves toward quasi-periodic behavior if simulations are run sufficiently long (Figure S7 in Supporting Information S1). The transition from aperiodic to quasi-periodic sequences occurs after 1,750 yr of simulation time, equivalent to 23 system-size earthquakes. This transition from aperiodic to periodic cycles implies that the initial complexity in the modeled earthquake sequences might represent a prolonged spin-up phase and that it might take tens of cycles to completely eliminate the effect of initial conditions.

The loss of complexity in the later stage of the simulation fundamentally questions the possibility of obtaining aperiodic earthquake sequences in 2D antiplane strain quasi-dynamic models. Previous studies that found non-characteristic (i.e., aperiodic) regimes often analyzed only a few hundred years, or equivalently, several system-size cycles, and it is not clear if the reported complexity persists over much longer time intervals (Cattania, 2019; Ito & Kaneko, 2023; Molina-Ormazabal et al., 2023; Nie & Barbot, 2022).

A more sustainable aperiodic behavior of modeled earthquake sequences may be achieved by explicitly accounting for the rough fault geometry (e.g., Cattania & Segall, 2021; Tal & Gabrieli, 2024), rather than approximating the effects of fault roughness (e.g., by applying spatially variable normal stress), as the topography of the fault surface perpetually introduces heterogeneity in stress to the system (Dunham et al., 2011; Fang & Dunham, 2013; Romanet et al., 2020). Whether complexity can be sustained beyond the timespans reported here in 2D antiplane strain quasi-dynamic simulations, and under what conditions, remains an open question for future research.

After the 1750-year transition to a periodic sequence (consisting of 2 repeating cycles), we observe repeating earthquakes ("repeaters"; Uchida & Bürgmann, 2019), such as the two unlabeled events preceding event 81 and event 88 in Figure 6. These repeaters occur at a depth of 11.36 km with a recurrence interval of 152 yr. The repeaters in this model show a significantly smaller slip (~0.3 m) than that expected from the creeping velocity at the VS area surrounding the repeater asperity (~5 m), similar to observations of natural repeaters (e.g., Chen et al., 2007; Nadeau & Johnson, 1998). These results suggest that caution should be exercised when using repeaters to infer local creep rates (Turner et al., 2024).

4.6. The Role of Fractal Heterogeneity in Slow Slip Generation

Within the parameter space explored, only models N7, N3R2D1, and N12R4D4 produce SSEs. In models N3R2D1 and N12R4D4, the emergence of SSEs is attributed to the presence of a conditionally stable VW patch near the base of the seismogenic zone, which is bounded by adjacent VS regions in the heterogeneous $(a - b)$ profiles version 2 and 4 (Figure S1 in Supporting Information S1). In both profiles, the VW patch is sufficiently large to permit nucleation and the adjacent VS regions are wide enough to arrest rupture propagation. This configuration, conditionally stable patches embedded within VS regions, is commonly adopted to generate SSEs in SEAS-type models (e.g., Wei et al., 2013).

The occurrence of SSEs is sensitive to the spatial distribution of fractal heterogeneities, particularly the size of the VS and VW patches, affecting their stability. For example, although models N3R1D1 and N3R2D1 share similar model parameters, only model N3R2D1 produces SSEs. This difference is likely due to the difference in the size of the VS patch near 12 km depth (0.2 km in N3R1D1 vs. 1 km in N3R2D1; Figure S1 in Supporting Information S1). In model N3R1D1, thinner VS zones are less effective as rupture barriers, while thicker VW patches promote faster slip (Lui & Lapusta, 2016; Wei & Shi, 2021). However, heterogeneous $(a - b)$ profiles alone cannot produce SSEs (e.g., model R4, Figure S9 in Supporting Information S1). Rather, the interplay between multiple heterogeneous parameters adds complexity to the system.

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In model N7, SSEs are facilitated by patches of low effective normal stress, typically ranging between 0.1 MPa and 1 MPa. Such low σ_n regions have been widely recognized as a mechanism for generating slow slip events in SEAS simulations (e.g., Y. Liu & Rice, 2007; Rubin, 2008; Wei et al., 2018). Their presence increases the critical nucleation length, allowing the patch to stay between nucleation and instability (Rubin, 2008).

5. Conclusions

We extensively explore the effects of various types of heterogeneities on generating non-characteristic earthquake cycles with considerable complexity in rupture size and hypocenter depth distributions in 2D quasi-dynamic rate-and-state friction simulations. Using the aging law, we find that a broad spectrum of fault slip, including both system-size and partial ruptures, occurs when multiple fractal heterogeneities in both stress and strength parameters are introduced conjointly. Creeping patches are critical for enabling depth-variable earthquake nucleation throughout the entire seismogenic zone due to elevated stressing rates at their margins. The presence of a low-rigidity fault zone adds little to system complexity compared to models with comparable W/L_∞ ratios but without a fault zone.

While model complexity and maximum W/L_∞ ratio exhibit moderate correlation, different models with similar W/L_∞ ratios show varying degrees of complexity. This result highlights that fault slip complexity can be highly model-dependent. As such, caution should be exercised when using the W/L_∞ ratio as a proxy for system complexity, particularly in heterogeneous models.

The most complex aging law model presented here (model N12R4D4) features system-size earthquakes with a range of hypocentral depths, which are consistently preceded by a cascade of partial ruptures, as well as shallow and deep slow slip events. In addition, this model reproduces other observed characteristics of natural faults, such as repeating earthquakes and aseismic transients preceding small-to-moderate-size earthquakes. However, even this most complex sequence transitions from aperiodic to periodic cycles after thousands of years, implying that the effects of initial conditions in cycle simulations may be more persistent than previously thought.

Finally, we compare earthquake sequences under aging and slip law assumptions. The slip law models produce shorter-duration quasi-periodic earthquake cycles, consisting of fewer earthquakes per cycle, smaller average static stress drops, and lower peak slip rates during system-size earthquakes.

The complex slip sequences analyzed in this study may serve as base models for a range of applications, as they resemble natural seismic and aseismic processes. For example, in the companion paper (Yun et al., 2025a), we use these models as reference slip histories for the 2019 M_w 7.1 Ridgecrest mainshock fault and investigate its triggering responses when subjected to external stress perturbations from the M_w 5.4 foreshock.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All input data required for reproducing all models are made available (Yun et al., 2025b). All simulations were conducted using the open-source software *Tandem* (Uphoff et al., 2022). An online documentation is available at https://tandem.readthedocs.io/en/latest/. We use dmay/seas-checkpoint branch (commit #1dc36db) for aging law simulations and jyun/state-law branch (commit #5d5c63f) for slip law simulations.

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